

Consideration of design viability

The proposed design was carefully reviewed based on several relevant extra- functional considerations; a judgment about design viability based on those considerations — the capacity of the proposed solution to address the problem — is clearly realistic and well supported with credible evidence.

During the considerations for different designs, we looked over a variety of different options when it came to aesthetic, maintainability, and extended use. We started by the four of us creating a list of 5 ideas each for ways of dropping the roses, including electromagnets, motors, strings/hooks, and fire. Then we narrowed the list of twenty down to 5 and created a decision matrix seen below. We then ranked each of the designs by 4 criteria, Simplicity 1-5, Cost 1-5, consistency 1-5, and remote controllability either true or false, being worth 10 points. We then ranked the designs and totalled the points, picking the highest number of points. After this we moved on to aesthetics and the actual details of the string design which we settled on. We decided that we needed to use hooks for ease of use, as we thought other ways would be to inconsistent. After reviewing the string we had to use we also decided that we wanted the string to be hidden, leading to the use of the kid wind pipe. We also determined that we would not be able to make the design remote controlled, leading to us having to consider other solutions, including a person hidden under the base of the set. We also decided to use air dry clay and green paint to make the design look more rose like.

Decision Matrix:

Rose Design	Simplicity (1-5 pts)	Cost (1-5 pts)	Consistency (1-5 pts)	Remote Controlled (T - 10 pts/F - 0)	Total
Electromagnets	1	3	5	T	19
String	3	4	4	T	21
Flame	2	1	2	T	15
Wind	3	2	1	T	16
Human Intervention	5	5	5	F	15